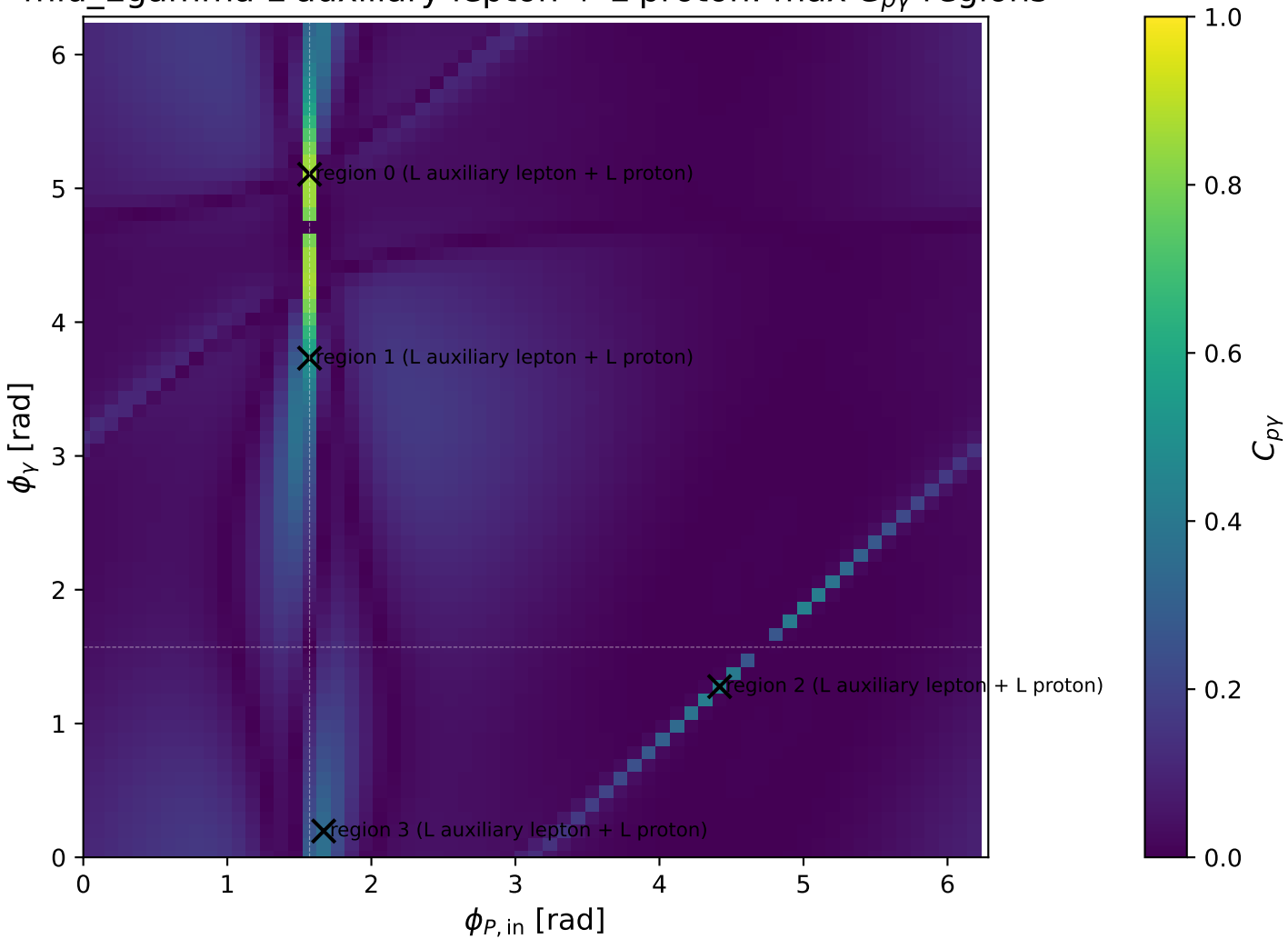
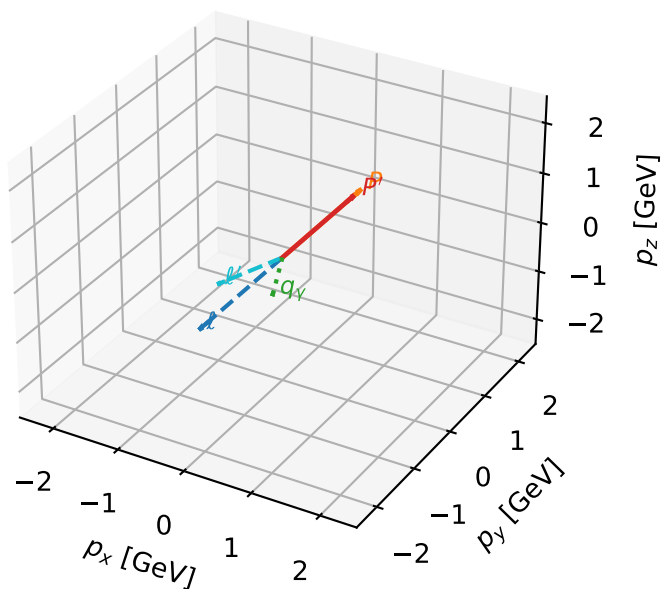


mid_Egamma L auxiliary lepton + L proton: max $C_{p\gamma}$ regions



L auxiliary lepton + L proton characteristic max $C_{p\gamma}$ configuration

3D momenta

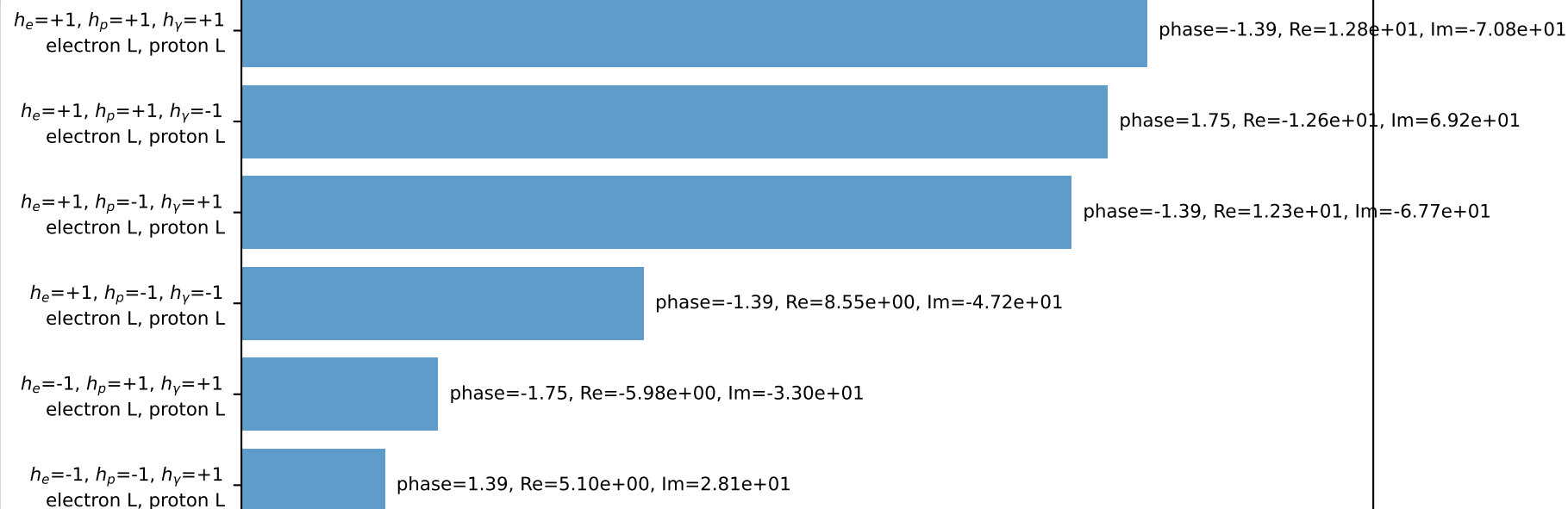
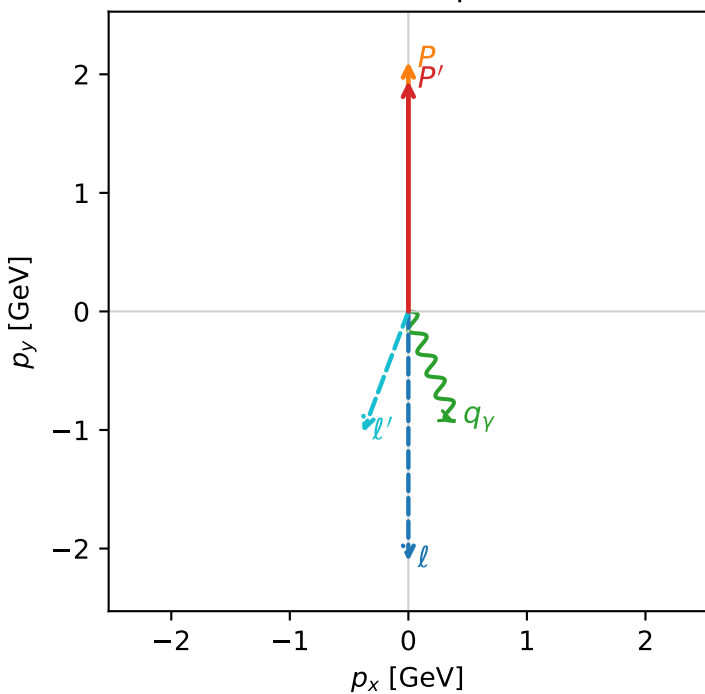


c_p_gamma_mid_egamma_ll_region_0 C_{p\gamma}=0.864314
 region: mid_Egamma_LL_region_0 final-pair delta_xy=2.74889 rad
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\ell\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.94473$
 $\theta_{in}=1.57079$, $\phi_\ell=4.71239$, $\phi_p=1.5708$
 $E_\gamma=1$, $\phi_\gamma=5.10509$
 $Q^2=0.598728$, $x_B=0.0703526$, $t=-0.00465004$, $W^2=8.7915$, $y=0.433407$
 $\theta(\ell', \gamma)=43.0493$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= 0.0000$
 ℓ' $E= 1.4794$ $p_x= -0.3827$ $p_y= -1.0209$ $p_z= 0.0000$
 P' $E= 2.1591$ $p_x= 0.0000$ $p_y= 1.9447$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.3827$ $p_y= -0.9239$ $p_z= 0.0000$

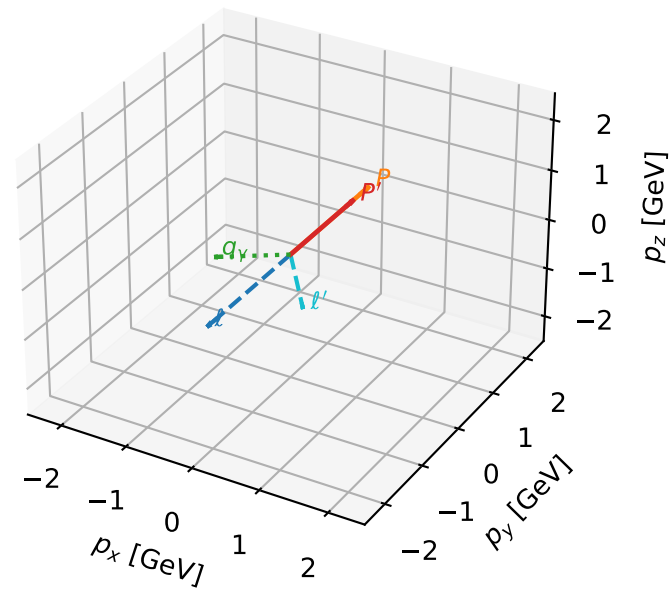
Transverse plane



ensemble-weighted final-state $|A|^2$ fraction

L auxiliary lepton + L proton characteristic max $C_{p\gamma}$ configuration

3D momenta

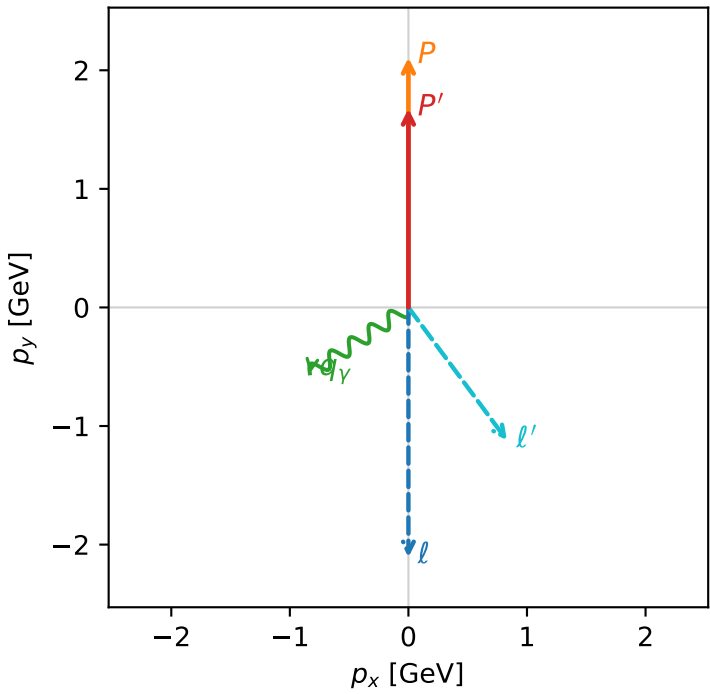


c_p_gamma_mid_egamma_ll_region_1 C_{p\gamma}=0.519984
region: mid_Egamma_LL_region_1 final-pair delta_xy=2.15984 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_l\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.67685$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_p=1.5708$
 $E_\gamma=1$, $\phi_\gamma=3.73064$
 $Q^2=1.28458$, $x_B=0.18376$, $t=-0.0368055$, $W^2=6.58579$, $y=0.356006$
 $\theta(l', \gamma)=92.8084$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= 0.0000$
 l' $E= 1.7171$ $p_x= 0.8315$ $p_y= -1.1213$ $p_z= 0.0000$
 P' $E= 1.9214$ $p_x= 0.0000$ $p_y= 1.6768$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.8315$ $p_y= -0.5556$ $p_z= 0.0000$

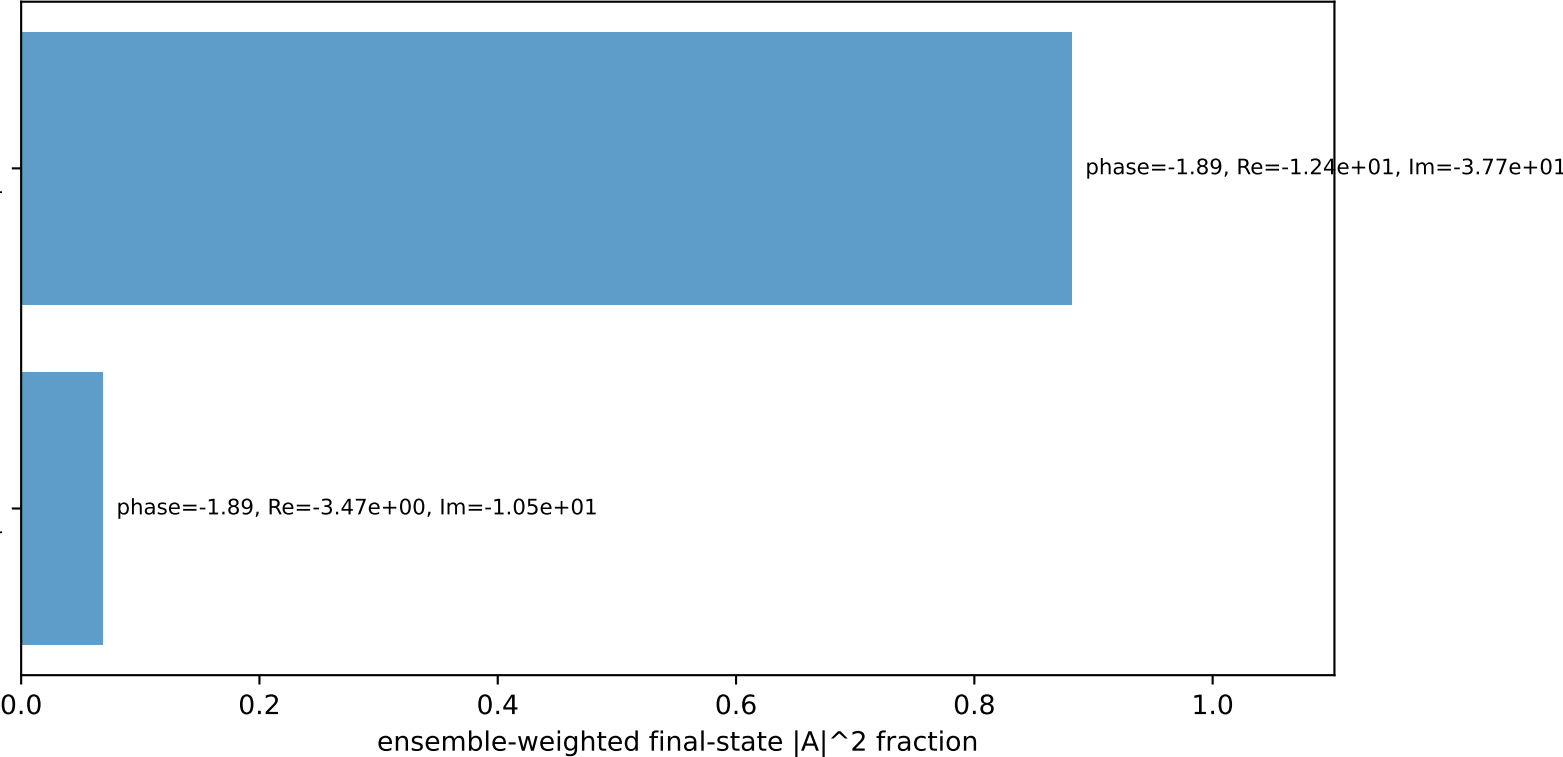
Transverse plane



$h_e=+1, h_p=-1, h_\gamma=+1$
electron L, proton L

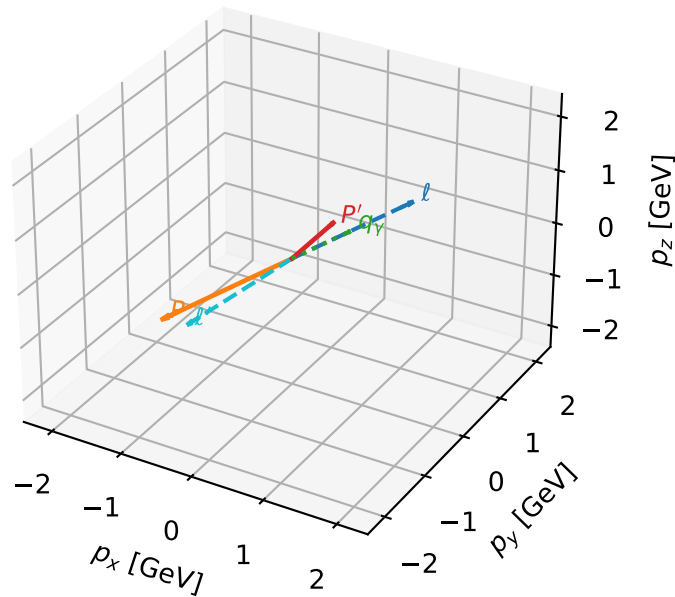
$h_e=+1, h_p=+1, h_\gamma=-1$
electron L, proton L

Leading initial-ensemble/final-state components (N=2, retained=95.0%)



L auxiliary lepton + L proton characteristic max $C_{p\gamma}$ configuration

3D momenta

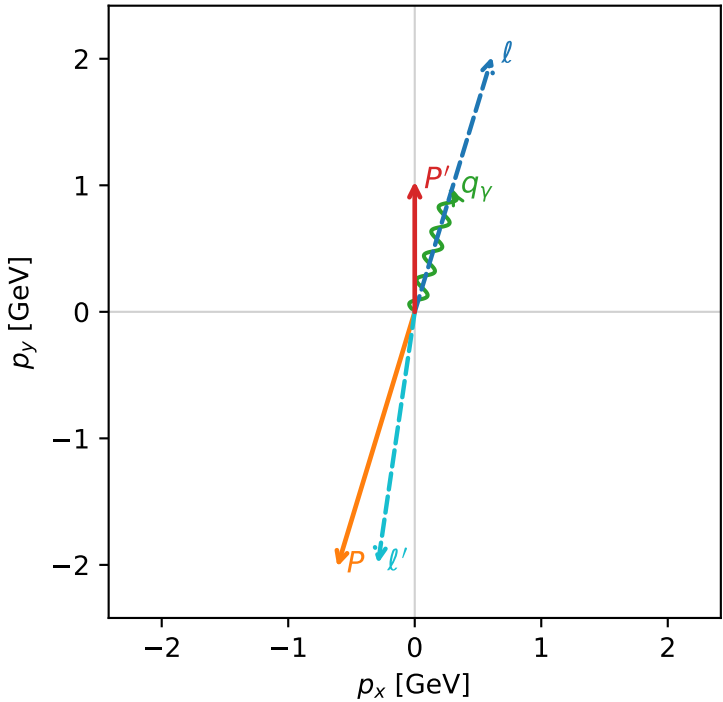


c_p_gamma_mid_egamma_ll_region_2 C_{p\gamma}=0.440568
region: mid_Egamma_LL_region_2 final-pair delta_xy=0.294524 rad
outgoing-state purity: 1
kinematic point: high s_high_theta_in_mid_Egamma
incoming state: $|L_\ell\rangle \otimes |L_p\rangle$

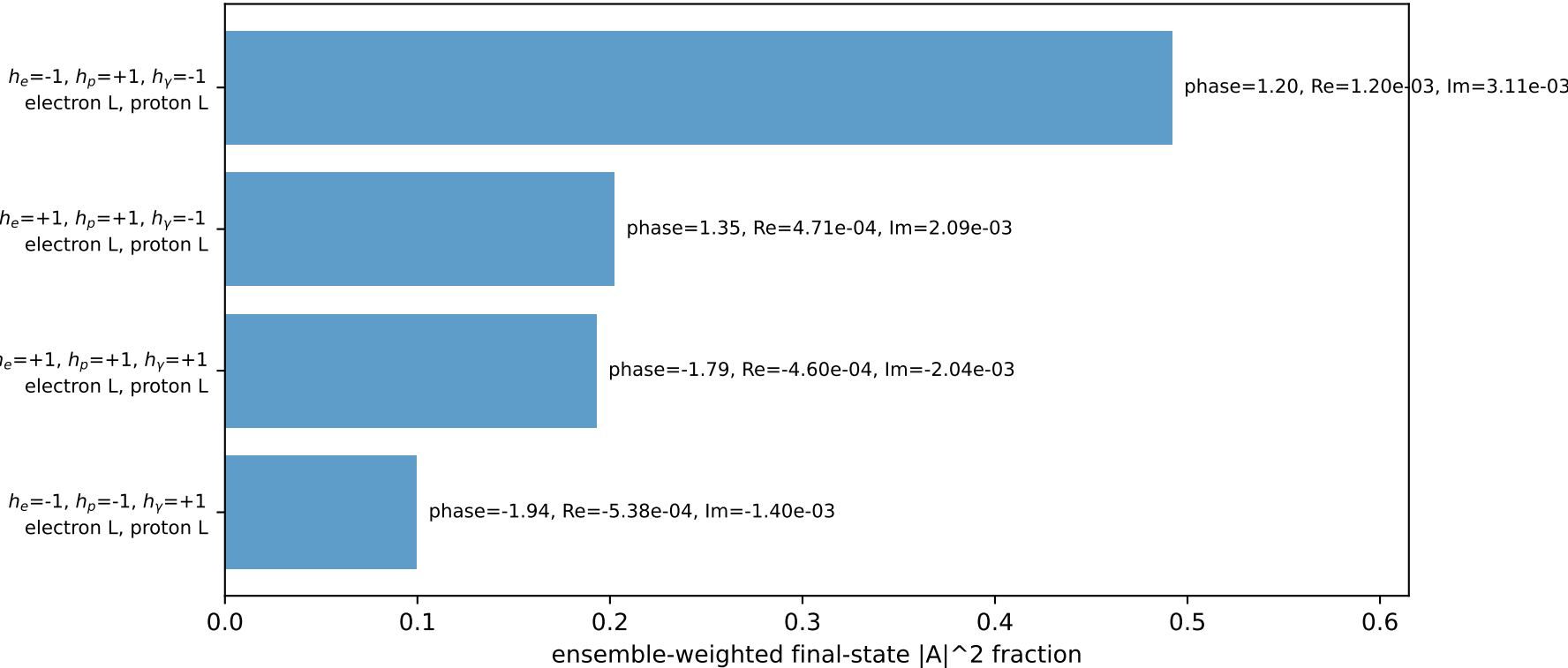
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.03133$
 $\theta_{in}=1.57079$, $\phi_\ell=1.27627$, $\phi_p=4.41786$
 $E_\gamma=1$, $\phi_\gamma=1.27627$
 $Q^2=16.8416$, $x_B=0.953877$, $t=-8.82945$, $W^2=1.69418$, $y=0.899162$
 $\theta(\ell', \gamma)=171.431$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 0.6116$ $p_y= 2.0162$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.6116$ $p_y= -2.0162$ $p_z= 0.0000$
 ℓ' $E= 2.2444$ $p_x= -0.2903$ $p_y= -1.9883$ $p_z= 0.0000$
 P' $E= 1.3941$ $p_x= 0.0000$ $p_y= 1.0313$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.2903$ $p_y= 0.9569$ $p_z= 0.0000$

Transverse plane

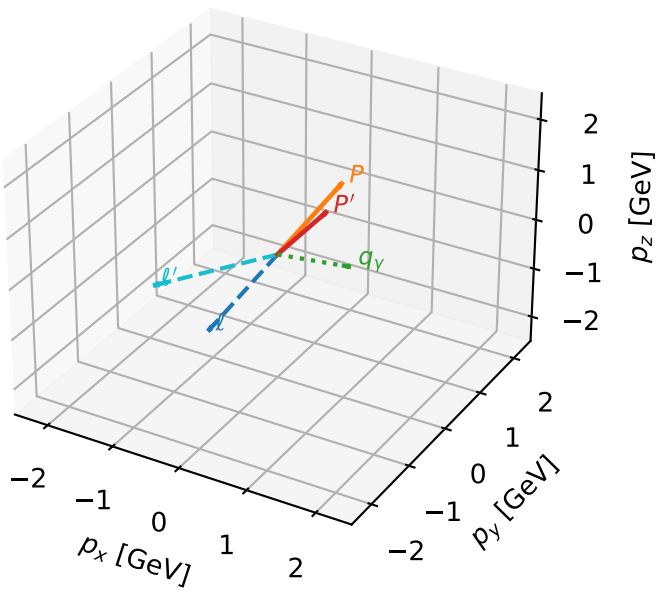


Leading initial-ensemble/final-state components (N=4, retained=98.7%)



L auxiliary lepton + L proton characteristic max $C_{p\gamma}$ configuration

3D momenta

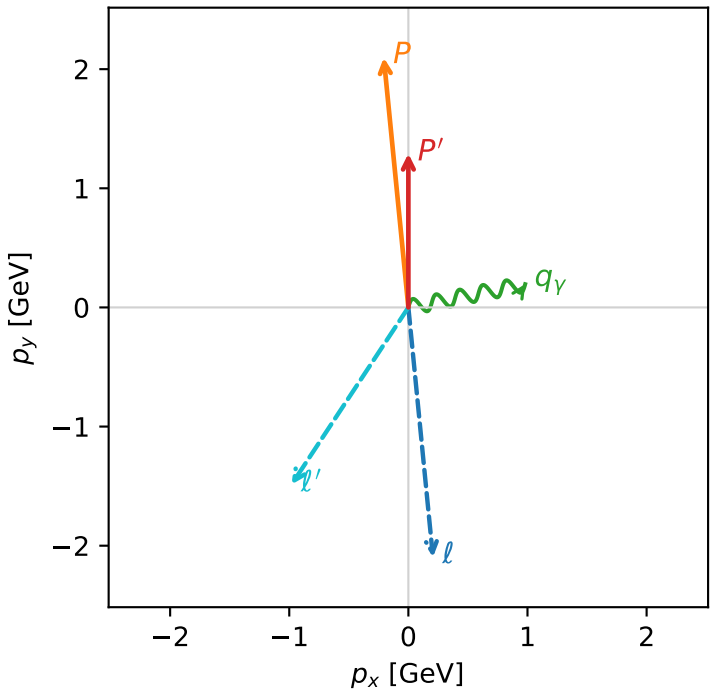


c_p_gamma_mid_egamma_ll_region_3 C_{p\gamma}=0.349177
region: mid_Egamma_LL_region_3 final-pair delta_xy=1.37445 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\ell\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.29138$
 $\theta_{in}=1.57079$, $\phi_\ell=4.81056$, $\phi_p=1.66897$
 $E_\gamma=1$, $\phi_\gamma=0.19635$
 $Q^2=1.69822$, $x_B=0.387144$, $t=-0.186941$, $W^2=3.56815$, $y=0.223392$
 $\theta(\ell', \gamma)=134.667$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 0.2065$ $p_y= -2.0968$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.2065$ $p_y= 2.0968$ $p_z= 0.0000$
 ℓ' $E= 2.0424$ $p_x= -0.9808$ $p_y= -1.4865$ $p_z= 0.0000$
 P' $E= 1.5961$ $p_x= 0.0000$ $p_y= 1.2914$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.9808$ $p_y= 0.1951$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=99.5%)

